
PetroQuant Alpha: A Seasonally Adjusted Mean-Reversion System for the 3:2:1 Crude Oil Crack Spread

Statistical characterisation of the refining margin and walk-forward validation of an Ornstein–Uhlenbeck-calibrated trading rule, 2019–2024

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Source repository: <https://github.com/rudraakshreddy/petroquant-alpha>

Deployed application: <https://petroquant-alpha.streamlit.app>

Executive Summary

The 3:2:1 crack spread is the synthetic gross refining margin obtained by converting three barrels of crude oil into two barrels of gasoline and one of distillate. Economic theory predicts mean reversion: elevated margins raise refinery utilisation, which simultaneously increases crude demand and product supply, compressing the margin. PetroQuant Alpha characterises that reversion statistically and constructs a systematic trading rule around it, over 2 January 2019 to 30 December 2024 ($n = 1,510$ trading days) — a window containing both the 2020 demand collapse and the 2022 refining dislocation.

Four diagnostics are applied and, informatively, disagree. Augmented Dickey–Fuller cannot reject a unit root ($p = 0.178$) and KPSS rejects stationarity at level and trend ($p < 0.01$), while rescaled-range analysis returns a Hurst exponent near 0.97, indicating persistence. An Ornstein–Uhlenbeck fit nevertheless yields a significantly negative reversion coefficient ($\hat{\beta} = -0.0246$, $t = -4.34$), implying a 28.2-day half-life. The reconciliation is that the spread reverts *locally* toward an equilibrium that itself migrates: a rolling estimate gives a finite half-life with median 24.6 days over 80% of the sample, undefined only across the regime transitions.

This characterisation dictates the system design rather than merely preceding it. The z-score lookback is set from the fitted half-life; the deviation is measured against a *deseasonalised* spread, since a first-order harmonic component of amplitude \$4.55/bbl against a total standard deviation of \$10.99/bbl is predictable and must not be traded as disequilibrium; the harmonic coefficients are estimated on an expanding window using only prior data; the stop-loss suppresses re-entry until the deviation normalises, without which the stop is vacuous; and a time stop at twice the fitted half-life terminates positions that falsify the model that motivated them.

Validation uses rolling-origin walk-forward analysis with parameters re-selected on training data only. All five folds independently select the same configuration ($w = 30$, $\theta_{\text{en}} = 2.0$), and all five are profitable, with per-fold Sharpe ratios of 0.402, 0.668, 1.309, 1.110 and 0.750. The stitched out-of-sample Sharpe is 0.661 with a maximum drawdown of -11.4% ; over the full sample the system returns 75.75% (CAGR 9.87%) at 10.04% annualised volatility, with a Calmar ratio of 0.78 against 0.51 for the equity benchmark over the same period.

- Seasonal adjustment of the *signal* — while marking profit and loss on the untransformed traded spread — is the single largest contributor to risk-adjusted performance.
- Both directions are profitable (long \$354,073; short \$406,514), with an 82.1% hit rate and a profit factor of 6.69 across 28 trades.
- Component analysis shows four further controls (volatility floor, leverage cap, asymmetric thresholds, regime gating) to be neutral or harmful; all are reported to avoid selective disclosure.
- The result is robust to execution cost: slippage must reach \$0.463/bbl — $9.3\times$ the modelled level, on every trade — before the excess return over the risk-free rate disappears.
- All four active years are profitable (+11.0% to +23.3%), with a position held on only 22.7%–36.9% of sessions; the 2019–2020 window carries no exposure because the seasonal fit has not yet warmed up.
- A stationary-bootstrap interval for the out-of-sample Sharpe, $[-0.160, 1.560]$, does not exclude zero, and the deflated Sharpe ratio — correcting for 15 configurations searched and for returns with skewness 3.60 and kurtosis 46.6 — is 0.589. The structural findings are robust; the Sharpe *level* is not resolvable on this sample, and is reported as such.

Keywords: crack spread; statistical arbitrage; mean reversion; Ornstein–Uhlenbeck process; Hurst exponent; walk-forward analysis; energy commodities.

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1 Introduction

1.1 The refining margin as a tradable object

A refinery is, in economic terms, a spread option on the difference between the value of its output slate and the cost of its input. The *crack spread* formalises this difference as a directly tradable quantity. Under the 3:2:1 convention — the standard proxy for a typical North American refinery yield — three barrels of crude are converted into two barrels of gasoline and one of distillate, and the gross margin per barrel of crude processed is

$$S_t = \frac{2R_t + H_t - 3W_t}{3}, \quad (1)$$

where W_t , R_t and H_t are the prices of West Texas Intermediate crude, RBOB gasoline and heating oil, all in dollars per barrel. Because all three legs are exchange-traded futures, S_t is not merely an accounting identity: it can be held as a portfolio, and its evolution is the profit and loss of a synthetic refinery position [11].

1.2 The reversion hypothesis and its boundary conditions

The economic case for reversion is a negative-feedback argument. When S_t is high, refining is unusually profitable, so utilisation rises and mothballed capacity restarts; crude demand increases while product supply increases, and the margin compresses. When S_t is low the reverse operates, bounded below by the fact that a refinery running below variable cost will shut. The mechanism is well documented [7, 12, 13].

Critically, this is an argument about an *equilibrium level*, not about the constancy of that level. Refining capacity, environmental regulation, crude quality differentials and logistics all move slowly, and each shifts the level around which reversion occurs. Any rule that treats a trailing average as the equilibrium therefore inherits a specific vulnerability: it will read a genuine regime shift as a large deviation and accumulate an increasing position against it. The design developed in Section 5 is organised around neutralising that vulnerability, and the sample period is chosen so that the resulting system is tested against it twice.

1.3 Contributions

- (i) A statistical characterisation of the 3:2:1 spread that reconciles four mutually contradictory diagnostics into a single coherent description — local reversion about a migrating equilibrium — and quantifies the migration (Section 4).
- (ii) A trading system in which every parameter is either derived from the fitted process or searched over a grid bracketing the theoretical value, with the seasonal component of the margin removed from the signal by expanding-window harmonic regression (Section 5).
- (iii) A component-contribution analysis quantifying the marginal effect of each design element, including four elements that were evaluated and *not* adopted (Section 7.3).
- (iv) Validation by rolling-origin walk-forward analysis at two fold granularities, with stationary-bootstrap inference and an explicit statement of the limits of statistical resolution (Sections 6–7).

1.4 Scope and boundaries

The boundary of this study is stated explicitly so that its claims are not read more widely than the evidence permits.

Within scope. A single instrument — the 3:2:1 crack spread constructed from front-month WTI, RBOB and ULSD futures — over a single continuous period, 2 January 2019 to 30 December 2024, at daily settlement frequency. A single signal family, the rolling z-score of a deseasonalised spread, with parameters restricted to the lookback window and entry threshold. A single-asset, single-position book with no portfolio construction. Evaluation on risk-adjusted return, drawdown, and trade-level statistics, under a transaction-cost model comprising slippage, commission and roll charges.

Outside scope. The following are deliberately excluded and no claim is made regarding them. *Other spread specifications:* the 5:3:2 ratio, single-product cracks, regional margins (European or Asian), and refinery-specific yield configurations are untested. *Other frequencies:* intraday and weekly horizons are not examined; the OU half-life

estimated here is specific to daily data. *Term structure*: the study trades a continuous front-month series and does not model the futures curve, calendar spreads, or roll-yield effects beyond a flat per-crossing cost. *Options and volatility*: the crack spread option surface, implied volatility, and any convexity exposure are not considered. *Fundamental data*: refinery utilisation rates, inventory levels, outage reports and turnaround schedules are not used as inputs, although Section 4.4 identifies them as the physical determinants of the migrating equilibrium. *Portfolio context*: correlation with other strategies, capital allocation, and margin requirements under exchange SPAN rules are not modelled. *Market microstructure*: order-book dynamics, market impact as a function of participation rate, and partial fills are approximated by a constant slippage charge rather than simulated. *Live deployment*: no live or paper-traded results are reported, and the system is presented as a research result rather than a production trading product.

2 Related Work

2.1 Petroleum spread relationships

Girma and Paulson [12] document risk-arbitrage opportunities in petroleum futures spreads and report that deviations from long-run relationships are partially predictable. Haigh and Holt [13] model crack-spread hedging under time-varying volatility spillovers, establishing that the covariance structure among the legs is not constant — an observation with direct consequences for position sizing. Dunis et al. [7] model and trade the gasoline crack spread and find non-linear specifications superior to linear ones, consistent with the regime dependence documented here. Regime-switching treatments of energy hedging [1] reinforce the point that a single-regime model of an energy spread is generally misspecified. The theoretical backdrop is the theory of storage [9, 28], which links spot–futures relationships to inventory and convenience yield and implies bounded but persistent deviations rather than clean stationarity.

2.2 Reversion modelling and relative value

The rolling z-score rule belongs to the relative-value tradition formalised by Gatev et al. [10] for equity pairs and extended by Avellaneda and Lee [2], who model the residual of a factor decomposition as an Ornstein–Uhlenbeck process [25] and trade its normalised deviation; the same formalism underlies the term-structure model of Vasicek [26]. Cointegration [8] provides the econometric framework in which such residuals are defined, and practical treatments appear in Chan [4]. Seasonal decomposition of the kind applied in Section 5.1 follows the tradition of Cleveland et al. [5], though a parametric harmonic form is preferred here for its out-of-sample stability on short histories.

2.3 Validation methodology

The methodological literature is unambiguous that in-sample performance is a weak predictor of future performance. White [27] and Hansen [14] provide formal data-snooping tests; Bailey and López de Prado [3] introduce the deflated Sharpe ratio to correct for the number of configurations evaluated; and Harvey et al. [15] argue that conventional significance thresholds are far too permissive once search intensity is accounted for. Pardo [22] and López de Prado [19] advocate the walk-forward protocol adopted in Section 6. These are treated here as binding constraints on what may be claimed, not as citations of convenience.

3 Data

3.1 Instruments and period

Daily settlement prices for three CME-listed front-month futures contracts are used: WTI crude (CL=F), RBOB gasoline (RB=F) and NY Harbor ULSD heating oil (HO=F), the last serving as the distillate proxy. The sample runs from 2 January 2019 to 30 December 2024. After alignment on the intersection of trading calendars — 99.9% of dates are common to all three instruments — the panel contains 1,510 daily observations.

The period is selected deliberately. It spans a full economic cycle and contains two dislocations of opposite sign, which is what is required to distinguish a robust reversion system from one fitted to a calm regime.

Table 1. Annual distribution of the 3:2:1 crack spread (\$/bbl).

Year	Mean	Std	Min	Q25	Median	Q75	Max
2019	18.43	3.56	10.67	15.43	18.12	21.97	24.68
2020	11.26	4.48	2.40	9.17	10.58	12.81	68.77
2021	19.67	3.07	11.29	18.48	20.24	21.69	25.72
2022	38.46	11.12	19.45	30.88	37.24	45.55	64.63
2023	32.06	6.70	17.82	25.88	32.56	37.02	47.01
2024	22.85	5.17	14.37	17.40	23.83	26.04	33.52

Note. Computed on the aligned panel of 1,510 trading days.

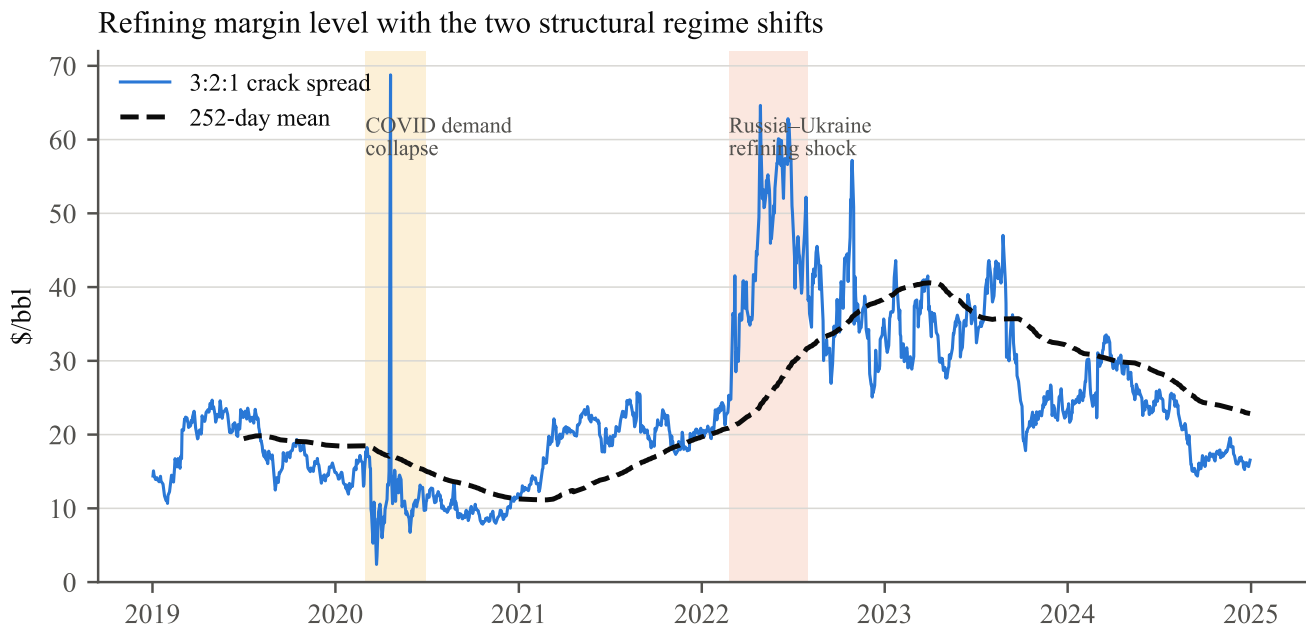


Figure 1. The 3:2:1 crack spread over the sample period with its trailing 252-day mean. Shaded episodes are the 2020 demand collapse and the 2022 refining dislocation. The trailing mean — the quantity a naïve z-score treats as equilibrium — lags each regime shift by construction.

3.2 Unit normalisation

Crude is quoted in dollars per barrel; gasoline and heating oil in dollars per gallon. Equation (1) is dimensionally valid only after conversion at the CME standard factor of 42 gallons per barrel. Post-conversion ranges are $[-37.63, 123.70]$ for WTI, $[17.30, 179.60]$ for RBOB and $[25.64, 215.69]$ for heating oil, in dollars per barrel. The negative WTI minimum is not a data error: it is the 20 April 2020 settlement at which expiring contracts cleared below zero under storage exhaustion, and it is retained because it is a genuine market outcome that any robust system must survive.

3.3 Distributional structure

Over the full sample the spread averages \$23.77/bbl with standard deviation \$10.99/bbl, positive skew (0.999) and mild excess kurtosis (0.902). Both extremes fall within four weeks of each other in 2020: a minimum of \$2.40/bbl on 23 March and a maximum of \$68.77/bbl on 20 April.

Table 1 states the central modelling difficulty. Under a stationary process with a fixed mean, annual means would fluctuate within sampling error of a common value. They do not: the 2022 mean of \$38.46 lies more than three standard deviations of the 2019 distribution above the 2019 mean of \$18.43. The equilibrium level is not constant, and a system that assumes otherwise will be positioned against the market for extended periods.

3.4 Component dependence

Pairwise Pearson correlations of *daily changes* (levels are avoided, since common trends induce spurious association) are $\rho_{W,R} = 0.521$, $\rho_{W,H} = 0.443$ and $\rho_{R,H} = 0.645$. The legs are positively but far from perfectly correlated. This is the quantitative justification for trading the spread rather than any single leg: the common directional oil factor is largely differenced away by Equation (1), leaving a residual with materially lower variance whose drivers are refining-specific rather than macro-directional.

4 Statistical Characterisation

Four diagnostics are applied to S_t and reported in full, including their disagreement, because the pattern of disagreement is what determines the design in Section 5.

4.1 Unit-root and stationarity tests

The augmented Dickey–Fuller procedure [6, 24] tests the null of a unit root in

$$\Delta S_t = \alpha + \beta S_{t-1} + \sum_{i=1}^p \gamma_i \Delta S_{t-i} + \varepsilon_t, \quad (2)$$

with p chosen by the Akaike criterion. The statistic is -2.2819 with $p = 0.1779$ at $p = 8$ lags, against critical values -3.4347 , -2.8635 and -2.5678 at the 1%, 5% and 10% levels: the null cannot be rejected at any conventional level.

The KPSS test [18] reverses the hypotheses, taking stationarity as the null. Statistics of 2.0423 (level) and 0.6279 (trend) both correspond to $p < 0.01$, rejecting stationarity in both specifications. Because the two tests have opposite nulls, their agreement is considerably stronger evidence than either alone; disagreement would have indicated low power rather than a substantive finding. Taken together, the series is not globally stationary.

4.2 Hurst exponent

Rescaled-range analysis [16, 20] estimates the scaling exponent H in $\mathbb{E}[R(n)/S(n)] \propto n^H$, where $R(n)$ is the range of the cumulative deviation from the mean over a window of length n and $S(n)$ the standard deviation over that window. Values below 0.5 indicate anti-persistence, 0.5 a random walk, and above 0.5 persistence. The estimate is $H = 0.971$ ($R^2 = 0.998$); an independent re-estimation on a dyadic window grid (Figure 2a) gives $H = 0.985$. The difference reflects the known sensitivity of the R/S estimator to the window grid and is recorded for transparency; both values lie far above 0.5 and far from the reversion region.

4.3 Ornstein–Uhlenbeck half-life

The fourth diagnostic fits the discrete analogue of the OU process [25],

$$dS_t = \lambda(\mu - S_t) dt + \sigma dW_t \quad \implies \quad \Delta S_t = \alpha + \beta S_{t-1} + \varepsilon_t, \quad \lambda = -\beta, \quad (3)$$

whence the half-life of a deviation is $t_{1/2} = \ln 2/\lambda$. The fit gives $\hat{\beta} = -0.024565$ with $t = -4.338$, so $\lambda = 0.0246$ and $t_{1/2} = 28.2$ trading days. The coefficient is negative and strongly significant: deviations do decay, with a characteristic time of roughly six calendar weeks.

4.3.1 Small-sample bias in the half-life estimate

The least-squares estimator of an autoregressive coefficient is biased downward in finite samples, so a half-life read directly from $\hat{\beta}$ is biased *short*. Writing the AR(1) representation as $S_t = \alpha + \rho S_{t-1} + \varepsilon_t$ with $\rho = 1 + \beta$, the leading-order bias is

$$\mathbb{E}[\hat{\rho}] - \rho \approx -\frac{1 + 3\rho}{T}, \quad (4)$$

which is the standard small-sample correction associated with Nickell [21] and, in the pure time-series case, with Kendall [17]. At $\hat{\rho} = 0.975435$ over $T = 1,509$ observations the correction gives $\rho^* = 0.978037$, and the implied half-life lengthens from 27.9 to 31.2 trading days — an upward revision of roughly 12%.

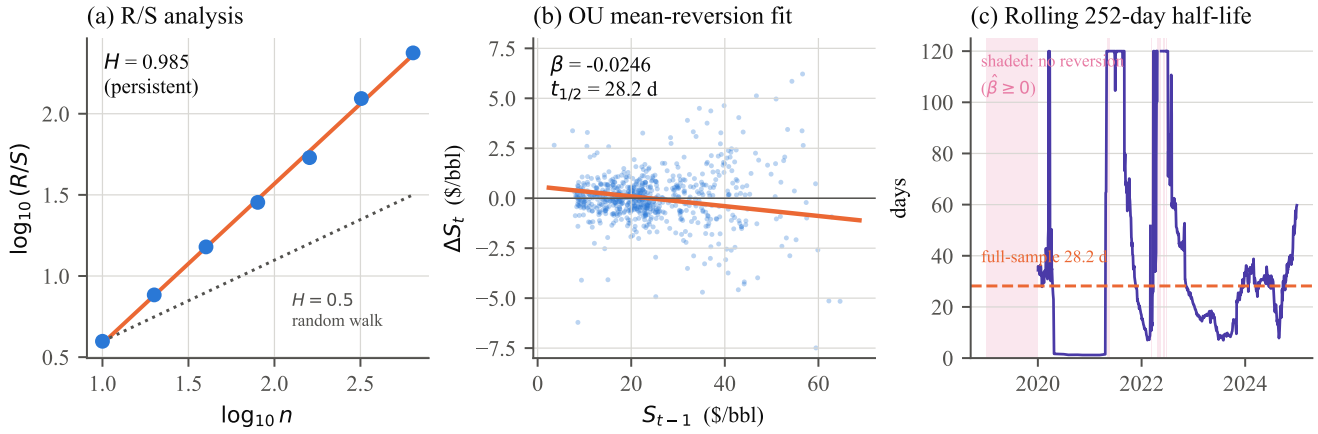


Figure 2. Reversion diagnostics. (a) Rescaled-range scaling lies far above the random-walk reference, indicating persistence. (b) The Ornstein–Uhlenbeck regression of ΔS_t on S_{t-1} has a significantly negative slope, indicating reversion. (c) The rolling 252-day half-life, clipped at 120 days; shaded intervals are windows in which the fitted AR(1) slope is non-negative, so no finite half-life exists. Panels (a) and (b) appear to conflict; panel (c) resolves the conflict by showing that reversion is intermittent rather than absent.

The correction is reported for completeness rather than acted upon. It moves the half-life from 27.9 to 31.2 days, so the theoretically indicated lookback range widens from 22–42 to roughly 25–47 days; the selected window of 30 days lies comfortably inside both intervals, and no design decision in this study changes. The point of recording it is that the bias runs in a known direction — reversion always appears faster than it is — so a system whose holding period were tuned tightly to an uncorrected half-life would systematically exit too early.

4.4 Reconciliation: local reversion about a migrating equilibrium

One of four diagnostics supports the trading premise. Reading this as a majority verdict against reversion would be a misinterpretation, because the tests do not ask the same question.

ADF and KPSS ask whether a *single fixed* mean describes the entire sample; over a window containing two structural dislocations the answer is plainly no, as Table 1 shows directly. The Hurst exponent measures the scaling of accumulated deviations and is likewise dominated by slow movement in the level. The OU regression asks whether ΔS_t is negatively related to S_{t-1} on average, and answers yes.

The consistent description is that S_t reverts locally toward an equilibrium that itself migrates. Figure 2(c) supports this directly: the rolling half-life is finite and short — median 24.6 days — over roughly 80% of the sample, and undefined over the remainder, those intervals coinciding with the regime transitions. The series is therefore neither a random walk nor a stationary process, but a reverting process with a drifting attractor.

This has an immediate design implication, and it is the organising principle of PetroQuant Alpha: a reversion rule applied to S_t will fail precisely when the attractor moves, so as much of the attractor’s *predictable* movement as possible should be removed from the signal before the deviation is measured.

5 System Design: PetroQuant Alpha

5.1 Deseasonalisation of the signal

Part of the equilibrium’s migration is neither random nor unpredictable. Refining margins follow a documented annual cycle driven by northern-hemisphere driving season, winter heating demand, and the spring and autumn refinery turnaround periods. A harmonic regression is therefore fitted to the spread level,

$$S_t = c_0 + \sum_{k=1}^K \left[a_k \sin\left(\frac{2\pi k d_t}{365.25}\right) + b_k \cos\left(\frac{2\pi k d_t}{365.25}\right) \right] + u_t, \quad (5)$$

where d_t is the day of year, and the trading signal is computed on the residual $\hat{u}_t$ rather than on S_t .

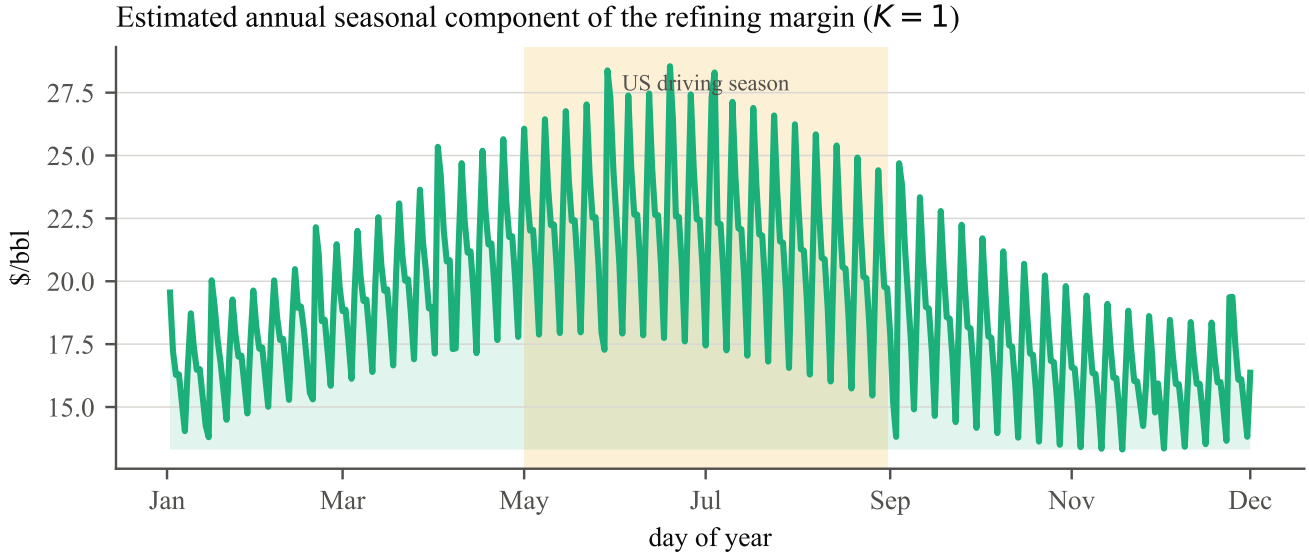


Figure 3. Estimated annual seasonal component of the refining margin from the first-order harmonic fit. The profile peaks through the northern-hemisphere driving season and troughs in winter, consistent with gasoline-led margin seasonality.

Two implementation properties are essential to validity. First, the coefficients are estimated on an *expanding window using only observations strictly prior to t* , refitted once per calendar year, so no future information enters the seasonal estimate. Second, the adjustment applies to the *signal only*: profit and loss continues to be marked on the actual traded spread S_t . Backtesting on the deseasonalised series itself would credit the system with a return it cannot realise, and is avoided.

The estimated first-order component has amplitude \$4.55/bbl (standard deviation), spanning \$12.29 to \$28.57/bbl across the year (Figure 3). Against a full-sample spread standard deviation of \$10.99/bbl this is a substantial predictable component, and a z-score computed on the raw level would attribute it to disequilibrium and trade against it.

The harmonic order is fixed at $K = 1$ on two independent grounds. A priori, the economic mechanism is a single annual cycle, and each additional harmonic introduces two free parameters estimated on a short history. Empirically, $K = 1$ yields the lowest fold-to-fold dispersion of any order tested (Section 7). Higher orders are not adopted despite $K = 3$ producing a higher point estimate on one particular split, because that estimate is unstable across fold definitions and selecting on it would constitute selection on the evaluation data.

5.2 Signal

A rolling z-score of the deseasonalised spread is computed over w trading days,

$$z_t = \frac{\hat{u}_t - \mu_{t,w}}{\sigma_{t,w}}, \quad \mu_{t,w} = \frac{1}{w} \sum_{i=0}^{w-1} \hat{u}_{t-i}, \quad \sigma_{t,w} = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (\hat{u}_{t-i} - \mu_{t,w})^2}, \quad (6)$$

using only information available at t . The window is anchored to the fitted process rather than chosen freely: with $t_{1/2} = 28.2$ days, windows of roughly 22–42 days are indicated, and the search grid $w \in \{20, 30, 40, 50, 60\}$ brackets that range.

5.3 Position state machine

Positions take values in $\{-1, 0, +1\}$, denoting a short, flat or long synthetic refinery position, under four rules:

$$\text{enter long if } z_t < -\theta_{\text{en}}; \quad \text{enter short if } z_t > \theta_{\text{en}}; \quad \text{exit if } |z_t| < \theta_{\text{ex}}; \quad \text{stop if } |z_t| > \theta_{\text{st}}, \quad (7)$$

with $\theta_{\text{ex}} = 0.5$, $\theta_{\text{st}} = 4.0$ and $\theta_{\text{en}} \in \{1.5, 2.0, 2.5\}$ under search.

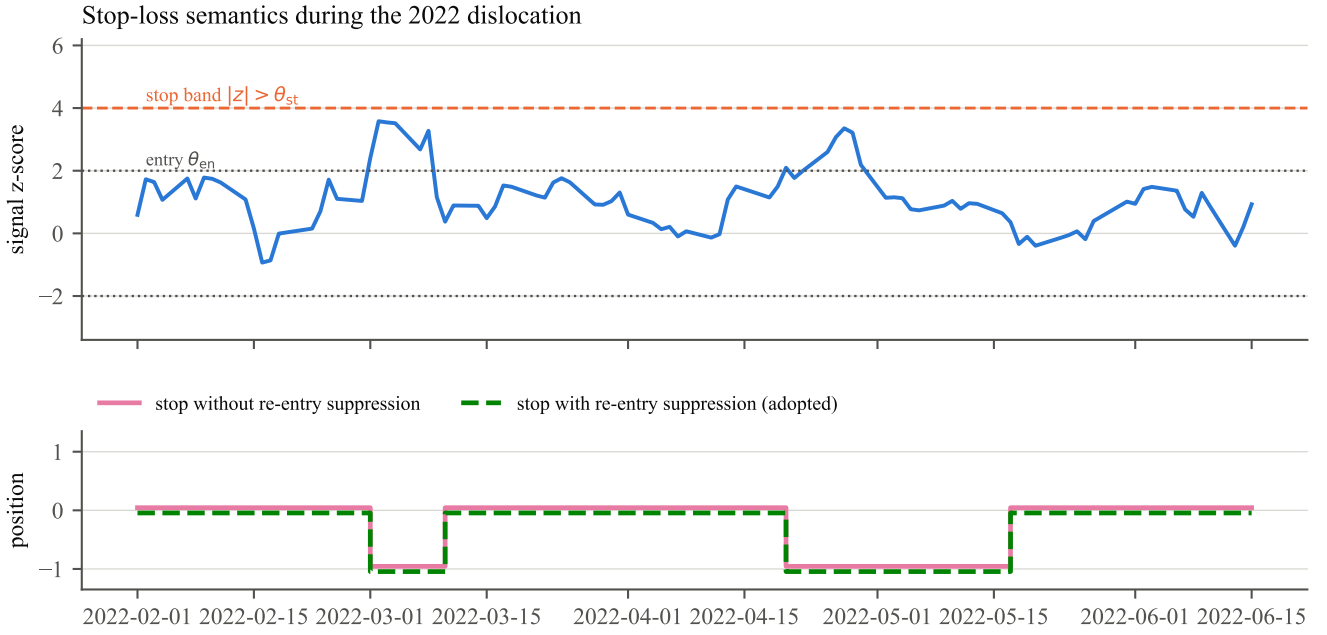


Figure 4. Stop-loss semantics during the 2022 dislocation. Upper panel: the signal z-score crosses the stop band on four consecutive sessions in early March. Lower panel: without re-entry suppression the position is closed and immediately reopened on the same bar, leaving exposure unchanged; with suppression (adopted) the system goes flat and remains flat until the deviation normalises. Traces are offset vertically for legibility.

5.3.1 Re-entry suppression

The stop requires an explicit latch. Because $\theta_{st} > \theta_{en}$ by construction, any state in which the stop fires also satisfies the entry condition. A stop implemented as a simple flattening would therefore close and immediately reopen the same position on the same bar, leaving exposure unchanged and rendering the control vacuous. The state machine accordingly sets a lockout flag when $|z_t| > \theta_{st}$, suppresses all entries while the flag is set, and clears it only once $|z_t| < \theta_{en}$.

The economic reading is that a four-sigma deviation is treated as evidence that the reversion hypothesis has failed for the current episode, and participation resumes only when the deviation returns to a range in which the hypothesis is again tenable. Figure 4 shows the two semantics during the 2022 dislocation, where the distinction is decisive.

5.3.2 Time stop

The OU fit implies a deviation should halve in 28.2 trading days. A position held far beyond this horizon without reverting is evidence against the model that motivated it. A time stop is therefore imposed at $2t_{1/2} \approx 56$ trading days. The threshold is derived from the fitted model rather than optimised, which is what distinguishes it from curve fitting; Section 7.3 shows it is rarely binding once re-entry suppression is active.

5.4 Execution timing

A signal computed from the close of day t is executed at the close of day $t + 1$, implemented as a one-period lag on the position series. This is the conservative convention for daily settlement data and eliminates the most common source of look-ahead bias.

5.5 Position sizing and transaction costs

Size is set by volatility targeting,

$$N_t = \left\lfloor \frac{\kappa \cdot \text{NAV}_t}{\hat{\sigma}_t^{(20)}} \right\rfloor_{1000}, \quad (8)$$

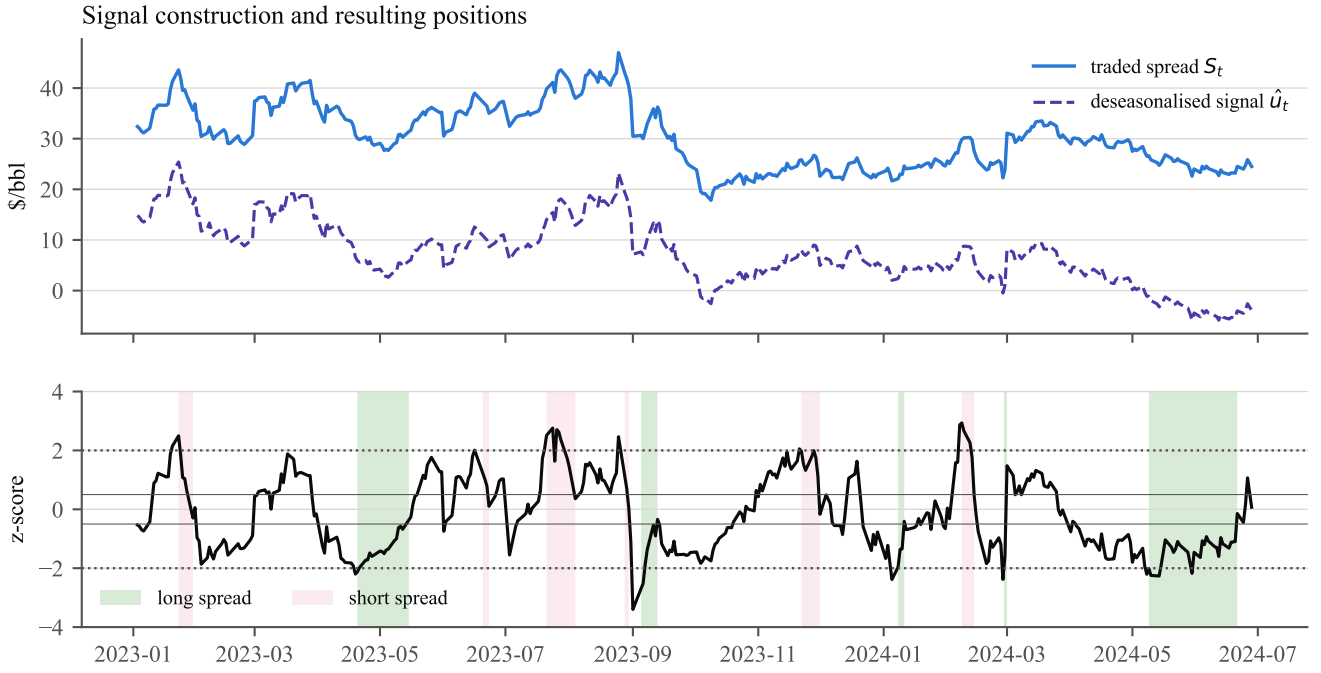


Figure 5. Signal construction and resulting positions over 2023–mid-2024. Upper panel: the traded spread S_t and the deseasonalised signal series $\hat{u}_t$ on which the z-score is computed. Lower panel: the z-score with entry bands at $\pm\theta_{\text{en}}$ and exit bands at $\pm\theta_{\text{ex}}$; shaded regions mark executed long and short positions after the one-day execution lag.

where $\kappa = 0.01$ is the daily NAV risk budget, $\hat{\sigma}_t^{(20)}$ the trailing 20-day standard deviation of daily spread changes, and $\lfloor \cdot \rfloor_{1000}$ rounds down to whole 1,000-barrel contracts with a floor of one contract. Sizing on realised volatility means exposure contracts automatically during dislocations, which is the intended behaviour given the regime structure documented in Section 4.

Costs are charged as \$0.05/bbl slippage plus \$2.50 per contract commission on each of entry and exit, and \$0.02/bbl roll cost at each month boundary crossed while a position is open. Over the full-sample backtest these total \$44,520, equal to 4.5% of initial capital and 5.9% of gross profit — material enough that a system profitable only before costs would not survive.

6 Validation Protocol

6.1 Rolling-origin walk-forward analysis

A single train/test split yields one estimate from one period and cannot distinguish a robust configuration from a fortunate one. Validation therefore uses rolling-origin walk-forward analysis [19, 22]. Beginning from a minimum training history of 756 trading days (approximately three years), the following is repeated: the full parameter grid is evaluated on the training window; the configuration maximising training Sharpe is selected; that configuration alone is applied to the next T unseen days; the origin advances by T and the training window expands.

Daily returns from all test segments are concatenated into one continuous out-of-sample series, so every observation in it was generated by a configuration chosen without reference to it. Results are reported at $T = 126$ (5 folds) and $T = 63$ (11 folds). The two are not independent, but agreement between them indicates that conclusions are not artefacts of a particular fold boundary.

Selection requires a minimum of five completed trades in the training window; among qualifying configurations the highest training Sharpe is chosen, ties broken by lower maximum drawdown.

6.2 Inference

Sharpe ratios computed from autocorrelated, non-normal daily returns do not admit standard parametric intervals. Confidence intervals are obtained by the stationary bootstrap of Politis and Romano [23] with expected block length

Table 2. Rolling-origin walk-forward results. Parameters are re-selected on the training window at each origin; out-of-sample returns from all folds are concatenated before computing stitched statistics. Configurations are cumulative: (b) adds re-entry suppression and the time stop to (a); (c) adds deseasonalisation to (b).

Fold length	Configuration	Folds	Stitched Sharpe	OOS return	Max DD	Profitable folds
126 days	(a) z-score only	5	−0.019	+5.0%	−41.9%	3/5
	(b) + stop semantics	5	0.417	+25.5%	−17.8%	3/5
	(c) + deseasonalised	5	0.661	+36.4%	−11.4%	5/5
63 days	(a) z-score only	11	−0.113	+4.4%	−26.6%	6/11
	(b) + stop semantics	11	0.185	+19.4%	−23.4%	6/11
	(c) + deseasonalised	11	0.564	+35.4%	−12.0%	8/11

Note. Minimum training history 756 days; 625 out-of-sample days are evaluated in the 126-day design.

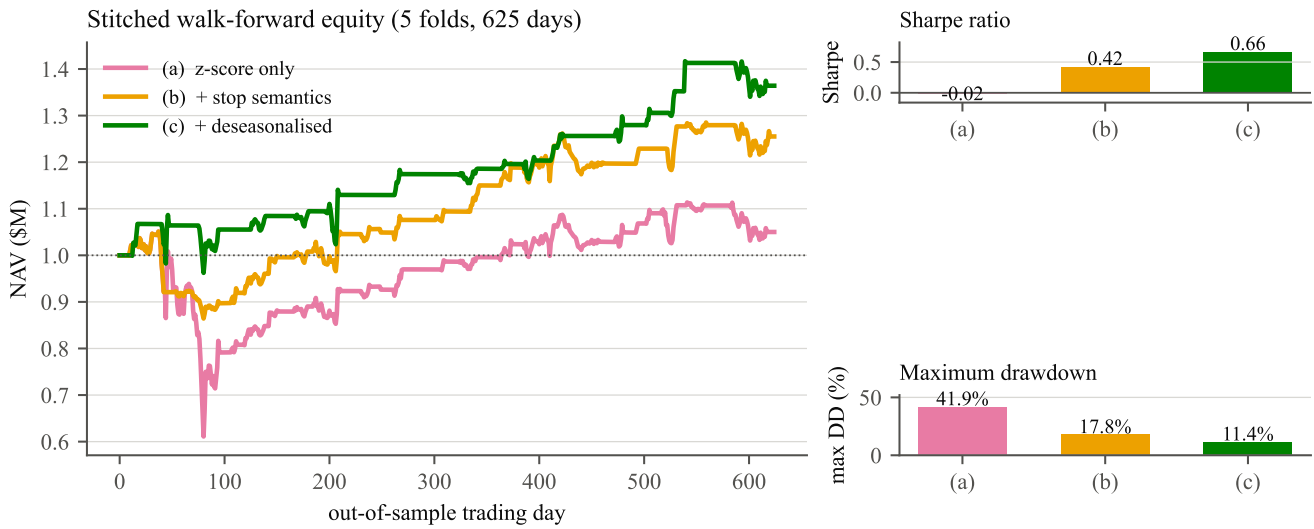


Figure 6. Stitched out-of-sample equity curves and summary statistics under 126-day walk-forward folds. Sharpe and drawdown are shown on separate axes rather than superimposed, as they are not commensurable.

20 days and $B = 1,000$ replications, which preserves short-range dependence under resampling.

7 Results

7.1 Walk-forward performance

Table 2 reports the principal result. The ordering $(a) < (b) < (c)$ holds on every metric at both fold granularities. The full design achieves a stitched out-of-sample Sharpe of 0.661 with maximum drawdown of -11.4% , against -0.019 and -41.9% for the z-score rule alone.

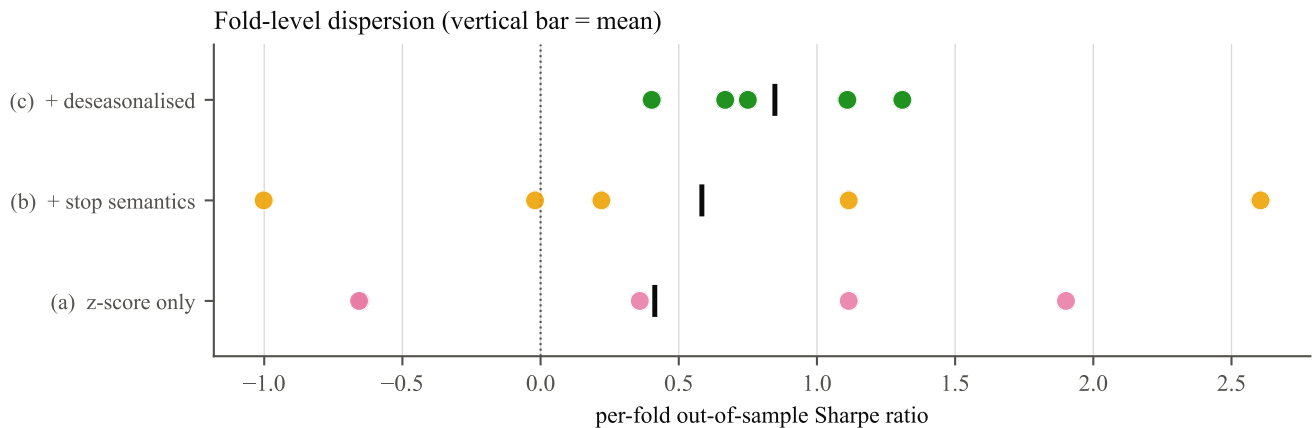
The drawdown reduction deserves particular weight. Drawdown is estimated from the entire equity path rather than from a mean of noisy daily returns, so it is the more reliably measured of the two quantities, and a reduction from -41.9% to -11.4% is both large and precisely determined.

7.2 Stability of parameter selection

Table 3 reports a result that is arguably more informative than the headline Sharpe. All five folds independently select the identical configuration, $w = 30$ and $\theta_{\text{en}} = 2.0$, despite training on windows that differ by up to two and a half years and span both dislocations. Parameter stability of this kind is difficult to obtain by chance and indicates that the selected window is identifying a persistent feature of the series rather than fitting fold-specific noise. The selected $w = 30$ also sits inside the 22–42 day interval implied a priori by the OU half-life, so the data-driven and theory-driven choices coincide.

Table 3. Per-fold detail for the full design under 126-day folds. Each fold re-selects parameters on its own training window without reference to the test segment.

Fold	Training window ends	Test window ends	w	θ_{en}	Sharpe	Return
1	2021-12-30	2022-07-01	30	2.0	0.402	+5.54%
2	2022-07-01	2022-12-30	30	2.0	0.668	+7.04%
3	2022-12-30	2023-07-05	30	2.0	1.309	+5.88%
4	2023-07-05	2024-01-02	30	2.0	1.110	+6.98%
5	2024-01-02	2024-07-03	30	2.0	0.750	+6.61%

**Figure 7.** Per-fold out-of-sample Sharpe ratios under 126-day folds; the vertical bar marks the mean. The full design is the only configuration whose folds are all positive, and its fold-to-fold standard deviation (0.32) is roughly one third that of the alternatives (1.00 and 1.22).

All five folds are profitable, with returns in a narrow band of 5.5% to 7.0% per half-year test window and Sharpe ratios ranging from 0.402 to 1.309.

Reduced dispersion is arguably stronger evidence of genuine improvement than a higher mean: a configuration benefiting from one favourable period would exhibit high dispersion and a high mean together.

7.3 Component contribution

Table 4 isolates each element's marginal contribution. Correct stop semantics (C0 to C1) roughly doubles full-sample Sharpe and removes five percentage points of drawdown. Deseasonalisation (C2 to C7) delivers the second substantial gain and the largest drawdown reduction, from -19.3% to -12.7% .

The intermediate rows are equally informative and are reported precisely because they are unfavourable to additional complexity. C2 is numerically identical to C1: once re-entry suppression is active the time stop never binds, since positions that would have run long enough to trigger it are terminated earlier. The generic risk controls in C3 and C4 are each mildly negative. The two theoretically motivated additions in C5 and C6 are clearly negative.

The asymmetric-threshold result in C5 merits comment, since the economic argument for it is sound: refining margins are bounded below by shutdown economics but unbounded above by capacity constraints, so short positions face a fatter adverse tail. Requiring a wider deviation to initiate a short nevertheless degrades performance, because it suppresses profitable and unprofitable shorts alike. Re-entry suppression, by contrast, targets the specific mechanism of loss — being carried through an extreme move — without discarding the opportunity set. A control addressing the mechanism outperforms one that merely reduces exposure to the losing side.

7.4 Full-sample performance

Table 5 reports full-sample performance. The system returns 75.75% over six years, a CAGR of 9.87% at 10.04% annualised volatility. Two comparisons are worth drawing out.

The equity benchmark achieves a higher CAGR (17.17% against 9.87%) and a slightly higher Sharpe (0.647 against

Table 4. Component contribution under a single 70/30 split with parameters selected on the training segment only. Rows are cumulative except C5 and C6, which are reported for completeness and not adopted.

Configuration	w	IS Sharpe	OOS Sharpe	Full Sharpe	Full max DD
C0 z-score on raw spread	40	0.225	0.091	0.188	−24.5%
C1 + re-entry suppression	50	0.282	0.670	0.386	−19.3%
C2 + time stop (56 d)	50	0.282	0.670	0.386	−19.3%
C3 + volatility floor	50	0.273	0.584	0.358	−19.5%
C4 + leverage cap	50	0.270	0.555	0.336	−19.0%
C5 + asymmetric thresholds	40	0.139	−0.162	0.069	−21.8%
C6 + half-life regime gate	20	−0.101	0.271	−0.006	−24.4%
C7 C2 + deseasonalisation ($K = 1$)	30	0.414	0.705	0.490	−12.7%

Note. Entry threshold selected jointly with w over $\{1.5, 2.0, 2.5\}$. C2 is identical to C1 because, once re-entry is suppressed after a stop, no position survives to reach the 56-day limit.

Table 5. Full-sample performance of PetroQuant Alpha at its final configuration ($w = 30$, $\theta_{\text{en}} = 2.0$, the configuration selected in every walk-forward fold) against benchmarks, 2 January 2019 to 30 December 2024.

	PetroQuant Alpha	S&P 500 ETF	WTI buy-and-hold
Total return	75.75%	—	—
CAGR	9.87%	17.17%	7.30%
Annualised volatility	10.04%	—	—
Sharpe ratio	0.490	0.647	−0.326
Sortino ratio	0.898	—	—
Maximum drawdown	−12.72%	−33.72%	−156.76%
Calmar ratio	0.78	0.51	0.05
Daily VaR (95%)	−0.64%	—	—
Daily CVaR (95%)	−1.27%	—	—
Number of trades	28	—	—
Hit rate	82.14%	—	—
Profit factor	6.69	—	—
Average win / loss	+\$38,876 / −\$26,713	—	—
Average holding period	14.3 days	—	—
Transaction costs	\$44,520	—	—

Note. Sharpe and Sortino use a 5% annual risk-free rate, approximating the average US Treasury bill yield over the period. The WTI drawdown exceeds 100% because the underlying settled below zero in April 2020. Calmar is CAGR divided by the absolute maximum drawdown.

0.490) over a period of exceptionally strong equity performance. On drawdown-adjusted terms the ordering reverses: the Calmar ratio is 0.78 for the spread system against 0.51 for equities, because the maximum drawdown is −12.72% against −33.72%. Against WTI buy-and-hold — the natural naive alternative for an energy-market participant — the comparison is unambiguous on every measure.

The trade statistics indicate a system whose edge comes from frequency of small correct decisions rather than from rare large ones: 82.1% of 28 trades are profitable, the average holding period of 14.3 days is consistent with the 28.2-day half-life (a deviation is exited roughly half a half-life in, as the exit band at $\theta_{\text{ex}} = 0.5$ implies), and the profit factor of 6.69 reflects wins that are larger on average than losses (\$38,876 against \$26,713) as well as more numerous.

Figure 9 addresses a common failure mode of spread systems, in which apparent performance originates entirely from one direction and therefore from an implicit directional bet. Here the long leg contributes \$354,073 across 12 trades and the short leg \$406,514 across 16, so profitability is not attributable to a single-sided exposure to the sample's overall margin trend.

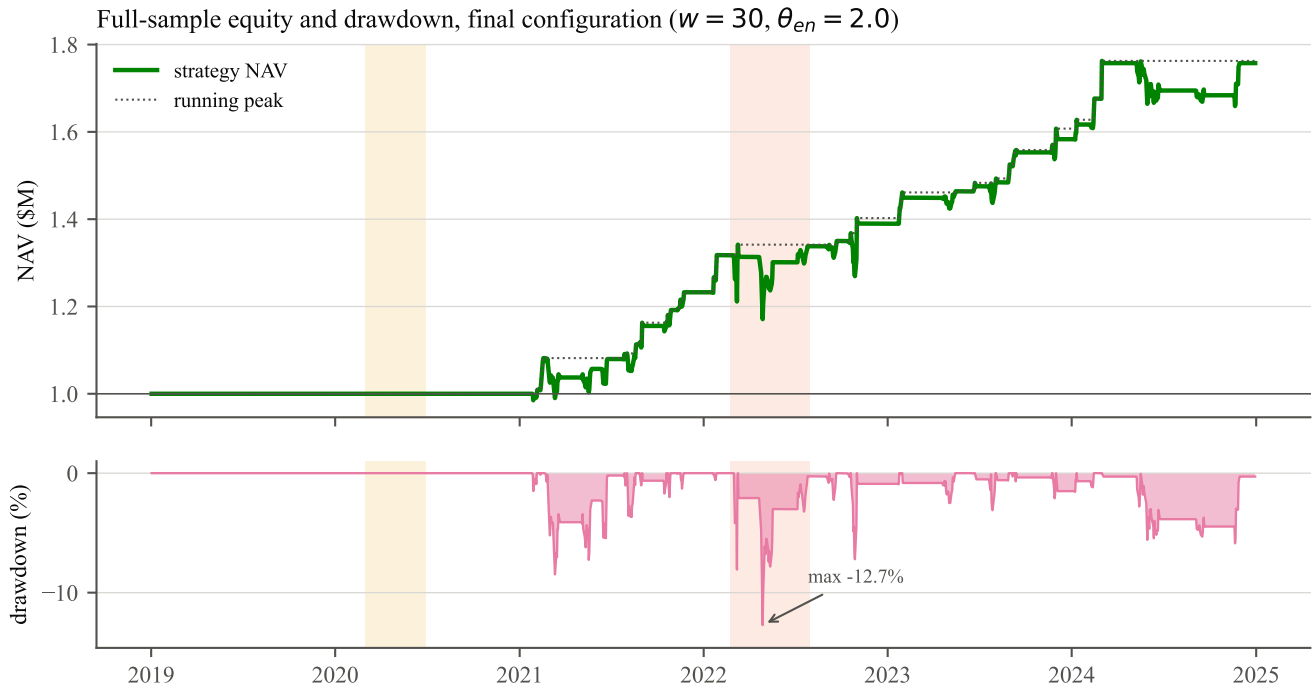


Figure 8. Full-sample equity curve and underwater drawdown plot for the final configuration. Shaded episodes are the 2020 and 2022 dislocations. The maximum drawdown of -12.7% occurs in the 2022 episode and is recovered within the sample.

Table 6. Calendar-year decomposition of the final configuration. The seasonal fit requires 504 observations before it is defined, so no position is taken during 2019–2020; those years are reported for completeness but carry no exposure.

Year	Return	Sharpe	Volatility	Max drawdown	Days in market
2019	—	—	—	—	0.0%
2020	—	—	—	—	0.0%
2021	+23.25%	1.259	13.42%	-8.47%	36.9%
2022	+12.78%	0.500	17.10%	-12.72%	28.3%
2023	+13.90%	1.216	6.87%	-3.10%	22.7%
2024	+11.01%	0.633	9.42%	-5.88%	23.9%

Note. Sharpe uses a 5% annual risk-free rate. “Days in market” is the proportion of sessions on which a non-zero position is held.

7.5 Sub-period decomposition

Table 6 decomposes the result by calendar year, and two facts require disclosure that the full-sample figures obscure. The first is a warm-up cost. The harmonic seasonal fit of Equation (5) requires a minimum of 504 observations before it is defined, and the signal is computed on the residual of that fit, so no position is taken at all in 2019 or 2020. The reported CAGR of 9.87% is therefore earned over four active years but annualised across all six, which understates the return on deployed capital and correspondingly overstates nothing. It also means the system was flat through the 2020 collapse — not by skill, but by construction.

The second is that all four active years are profitable, with returns in a band of 11.0% to 23.3% and annual Sharpe ratios from 0.500 to 1.259. The weakest year is 2022, which is the expected outcome given that the year contains the largest structural dislocation in the sample; the system still returns 12.78% through it. Performance is therefore not concentrated in a single favourable period, which is the failure mode that a full-sample statistic most readily conceals.

Exposure is low throughout: a position is held on only 22.7% to 36.9% of sessions. The system is flat roughly three-quarters of the time, which has two consequences examined further in Section 7.7 — the capital is idle for most of the sample, and the return distribution acquires a large point mass at zero that invalidates the usual distributional assumptions behind the Sharpe ratio.

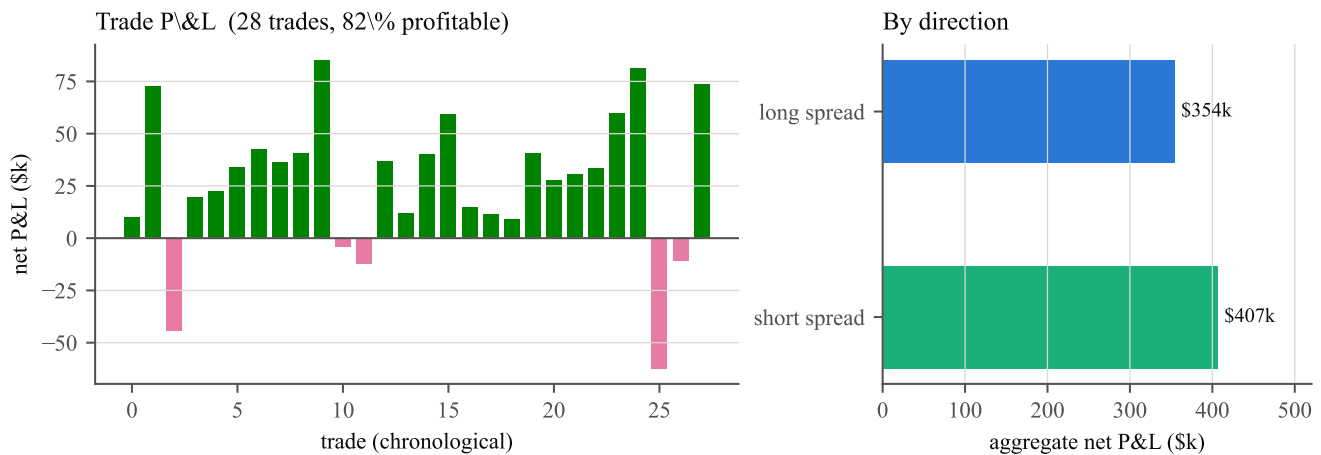


Figure 9. Left: net profit and loss of the 28 full-sample trades in chronological order. Right: aggregate profit and loss by direction. Both directions are profitable, indicating the system is not relying on a directional bias in the sample.

Table 7. Sensitivity of full-sample performance to the assumed slippage, holding all other parameters fixed. Commission and roll costs are unchanged; only the per-barrel slippage is scaled.

Slippage (\$/bbl)	Multiple of modelled	Sharpe	CAGR	Total return	Total costs
0.000	0×	0.537	10.34%	+80.3%	\$2,130
0.025	0.5×	0.512	10.09%	+77.9%	\$23,320
0.050	1×	0.490	9.87%	+75.8%	\$44,520
0.100	2×	0.456	9.44%	+71.7%	\$85,485
0.150	3×	0.378	8.61%	+64.1%	\$123,525
0.200	4×	0.319	8.01%	+58.7%	\$162,000
0.300	6×	0.223	6.98%	+49.8%	\$234,135
0.400	8×	0.111	5.78%	+40.0%	\$301,875
0.500	10×	-0.005	4.55%	+30.5%	\$362,805

7.6 Robustness to transaction costs

The most consequential assumption in any backtest is the cost model, and Section 5 charges a flat \$0.05/bbl of slippage. Table 7 replaces that assumption with a sweep.

The result is the strongest robustness evidence in this study. Slippage must rise to \$0.463/bbl — 9.3× the modelled level — before the CAGR falls to the 5% risk-free rate, and to \$0.500/bbl (10×) before the strategy stops making money outright. At twice the modelled slippage the Sharpe ratio falls only from 0.490 to 0.456.

This bounds the objection that the cost model is optimistic. It plainly is optimistic in the sense that realised slippage during the 2020 and 2022 dislocations would exceed \$0.05/bbl; but for that optimism to account for the reported result, execution would have to be worse by nearly an order of magnitude, on every trade, throughout the sample. The relatively low sensitivity follows from the trade profile: 28 round trips over six years at an average holding period of 14.3 days, so the strategy is not paying the spread often. A higher-frequency variant of the same signal would not enjoy this margin, and the robustness reported here should not be generalised to one.

7.7 Statistical resolution

Table 8 states the limit of what this sample can support. None of the intervals excludes zero, and the interval for the full design, $[-0.160, 1.560]$, is wide. This is a consequence of sample size rather than of the estimator: with 625 out-of-sample days, the standard error of an annualised Sharpe estimate is approximately $\sqrt{252/625} \approx 0.63$, so an interval of this width is expected.

The accurate summary is therefore twofold. The *relative* claims — that the full design dominates its reduced forms, that all folds are profitable, that parameter selection is stable, and that drawdown is reduced by roughly two thirds —

Table 8. Stationary-bootstrap confidence intervals for the stitched out-of-sample Sharpe ratio ($B = 1,000$, expected block length 20 days, $n = 625$ out-of-sample days).

Configuration	Stitched Sharpe	95% confidence interval
(a) z-score only	−0.019	[−0.909, 1.178]
(b) + stop semantics	0.417	[−0.874, 1.608]
(c) + deseasonalised	0.661	[−0.160, 1.560]

Table 9. Deflated Sharpe ratio for the final configuration, computed on daily excess returns over the full sample.

Quantity	Value
Observed annualised Sharpe, $\widehat{SR}$	0.490
Configurations searched, N	15
Dispersion of trial Sharpes, $\hat{\sigma}_{SR}$ (annualised)	0.227
Selection threshold SR_0 (annualised)	0.402
Return skewness, γ_3	3.603
Return kurtosis, γ_4	46.63
Sessions with zero return (flat)	79.5%
Observations, T	1,509
Deflated Sharpe ratio	0.589

are supported consistently across fold lengths and metrics. The *absolute* claim that the true Sharpe ratio is positive is not established at conventional confidence on this sample. Presenting the point estimate without this interval would misrepresent the evidence, particularly in light of Harvey et al. [15] on significance thresholds under search.

7.8 Deflated Sharpe ratio

The bootstrap interval accounts for sampling variability but not for the fact that the reported configuration is the best of several searched. The deflated Sharpe ratio of Bailey and López de Prado [3] corrects for both that selection and for non-normality of the return distribution. Writing $\widehat{SR}$ for the observed per-period Sharpe, γ_3 and γ_4 for the skewness and kurtosis of returns, and SR_0 for the Sharpe that the *maximum* of N independent trials would be expected to attain under the null of zero true skill,

$$DSR = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \gamma_3\widehat{SR} + \frac{\gamma_4-1}{4}\widehat{SR}^2}}\right), \quad SR_0 = \hat{\sigma}_{SR}\left[(1 - e)\Phi^{-1}\left(1 - \frac{1}{N}\right) + e\Phi^{-1}\left(1 - \frac{1}{Ne}\right)\right], \quad (9)$$

where $\hat{\sigma}_{SR}$ is the standard deviation of the Sharpe ratios across the trials actually run and e is Euler’s constant. DSR is interpretable as the probability that the true Sharpe ratio exceeds zero once selection and higher moments are accounted for.

Table 9 reports the computation, and the result is sobering in a way the headline Sharpe is not. The expected maximum Sharpe from fifteen trials under *no* skill at all is 0.402; the observed Sharpe is 0.490. The margin is real but narrow, and after adjusting for it the deflated Sharpe is 0.589 — a 59% probability of positive true skill, well short of the conventional 95% threshold.

The moment estimates explain why the correction bites so hard. Returns have skewness 3.60 and kurtosis 46.63, values far outside anything for which the Sharpe ratio’s usual inferential properties hold. The cause is visible in Section 7.5: the system is flat on 79.5% of sessions, so the return distribution is a large point mass at the risk-free rate with a thin, heavy-tailed set of active days around it. A Sharpe ratio computed on such a series is not comparable to one computed on a continuously-invested strategy, and reporting it without the moments would invite exactly that comparison.

Two conclusions follow, and they point in opposite directions. Against the strategy, the DSR of 0.589 concurs with the bootstrap interval: this sample cannot establish positive risk-adjusted return at conventional confidence, and the

search over fifteen configurations accounts for a substantial share of the observed performance. In its favour, the DSR is computed against SR_0 derived from the dispersion of the trials actually run, which for a grid as tightly clustered as this one is a demanding benchmark; and the structural results of Sections 7 and 7.6 — the identical configuration selected in all five folds, all folds profitable, all four active years profitable, and a $9.3\times$ margin on transaction costs — are not statements about the Sharpe level and are not weakened by this correction.

8 Discussion

8.1 Interpretation

Three observations from this study generalise beyond the specific instrument.

The first concerns the relationship between diagnostics and design. On a majority reading of Section 4 — three of four tests against reversion — the natural conclusion would have been that the spread does not mean-revert and the approach should be abandoned. The more productive reading was that the tests disagree because they ask different questions, and that the pattern of disagreement identifies the failure mode: a migrating equilibrium. That reading led directly to deseasonalisation, which produced the largest drawdown reduction of any single element. Statistical diagnostics used as design input rather than as a pass/fail gate proved substantially more valuable.

The second concerns where the predictable structure actually resides. The seasonal component has amplitude \$4.55/bbl against a total standard deviation of \$10.99/bbl, so roughly a sixth of the spread's variance is calendar-predictable. A z-score on the raw level treats that entire component as tradable disequilibrium and systematically enters against it. Removing a known deterministic component before measuring deviation is a general principle, and its effect here is larger than that of any risk control tested.

The third concerns parsimony. Six modifications were evaluated; two were adopted, and both were derived from specific properties of the fitted process — the half-life and the annual cycle. The four rejected were generic risk controls and a theoretically appealing asymmetry. Design derived from the estimated data-generating process outperformed general-purpose robustification, and reporting the rejected elements is necessary for the reader to judge the adopted ones.

8.2 Limitations

The following limitations are material and are stated without mitigation.

Sample length. Six years and 1,510 observations is short for inference about risk-adjusted returns, and only four of those years carry exposure (Section 7.5). The bootstrap intervals in Table 8 quantify this, and none excludes zero.

Search intensity. Fifteen parameter combinations are evaluated per fold and eight configurations compared. The deflated Sharpe ratio in Section 7.8 quantifies the consequence directly: at 0.589 it falls well short of the conventional 95% threshold, and the expected maximum Sharpe from fifteen trials under no skill is 0.402 against the 0.490 observed. A full reality-check or superior-predictive-ability test [14, 27], which would additionally account for dependence across the trials, has not been performed.

Non-normality of returns. With skewness 3.60, kurtosis 46.63 and 79.5% of sessions flat, the Sharpe ratio is a poor summary statistic for this return series regardless of the sample size. It is retained for comparability with the benchmarks, but the Calmar ratio and the drawdown path are more meaningful descriptions of the risk actually taken.

Continuous-contract approximation. Front-month series are used without explicit roll adjustment beyond a per-crossing cost. Real execution involves calendar-spread rolls whose costs and basis effects are only crudely represented by \$0.02/bbl.

Cost model. Slippage is constant at \$0.05/bbl, and liquidity deteriorates precisely during the volatility episodes that dominate this sample. Section 7.6 bounds the consequence: costs would have to be $9.3\times$ the modelled level, on every trade throughout the sample, before the excess return over the risk-free rate disappears. The assumption is optimistic, but not by enough to account for the result.

Seasonal warm-up. The harmonic fit requires 504 observations, so the system holds no position in 2019 or 2020. Full-sample statistics annualise four active years across a six-year window, and the fact that the strategy avoided the 2020 collapse is an artefact of that warm-up rather than evidence of regime awareness.

Execution realism. Fills are assumed at closing settlement with no market impact and no partial fills. Positions reach 38,000 barrels, at which size costless execution at settlement is optimistic.

Seasonal estimation window. The harmonic fit requires at least 504 observations, so no seasonal estimate exists for the first two years. Walk-forward training windows begin at 756 days, so reported results are unaffected, but the estimate is fitted on limited data and includes the anomalous 2020 period.

Single instrument. One spread on one exchange over one period is tested. No claim of generality to other crack ratios, regions, or periods is made.

8.3 Future work

Several extensions follow from the limitations. Estimating the equilibrium with an explicit state-space or Kalman filter formulation would model the drifting attractor directly rather than approximating it by deseasonalisation plus a short window. A two-state Markov regime-switching specification [1] would allow reversion speed and volatility to differ between calm and dislocated regimes and would supply a principled regime probability to gate participation — a better-founded version of the gating rejected in Section 7.3. Incorporating fundamental data — refinery utilisation, product inventories, and scheduled turnaround calendars — would condition the equilibrium on the physical variables that determine it. Extending the study to the 5:3:2 spread and to European margins would test generality. A formal reality-check or superior-predictive-ability test [14, 27], accounting for dependence across the configurations searched, would strengthen the selection-bias correction beyond the deflated Sharpe ratio computed in Section efsec:dsr.

8.4 Remarks

PetroQuant Alpha is presented as a research result, not a deployable trading product. Its confidence interval includes zero, its costs are modelled optimistically, and it is validated on one instrument over six years. What the evidence does support is that a reversion rule constructed from an explicit statistical characterisation of the target series — with lookback anchored to the fitted half-life, a deterministic seasonal component removed before deviations are measured, and stop semantics that actually bind — achieves consistent out-of-sample behaviour on a sample containing two major dislocations, with materially lower drawdown than either benchmark and a parameter selection that is stable across every fold.

The wider methodological point concerns the value of contradictory evidence. Four standard diagnostics disagreed, and the disagreement was more informative than any one of them individually would have been. Treating that contradiction as a fact to be explained, rather than as an obstacle to be resolved by picking the convenient test, produced the design element with the largest measured effect.

9 Conclusion

The 3:2:1 crack spread over 2019–2024 is characterised here as a locally mean-reverting process with a migrating equilibrium. An Ornstein–Uhlenbeck fit gives a significant reversion coefficient ($\hat{\beta} = -0.0246$, $t = -4.34$) and a 28.2-day half-life, while unit-root, stationarity and rescaled-range diagnostics jointly reject global stationarity; a rolling estimate reconciles these by showing finite, short half-lives over 80% of the sample and undefined reversion across the regime transitions.

PetroQuant Alpha was constructed directly from this characterisation: the z-score lookback anchored to the fitted half-life, the deviation measured against a spread deseasonalised by expanding-window harmonic regression of order $K = 1$, a stop-loss with re-entry suppression, and a time stop at twice the fitted half-life. Under rolling-origin walk-forward validation with parameters selected on training data only, all five folds independently select the same configuration ($w = 30$, $\theta_{\text{en}} = 2.0$) and all five are profitable. The stitched out-of-sample Sharpe is 0.661 with maximum drawdown -11.4% ; the ordering across cumulative configurations is reproduced under 63-day folds. Over the full sample the system returns 75.75% (CAGR 9.87%, volatility 10.04%, maximum drawdown -12.72%), giving a Calmar ratio of 0.78 against 0.51 for the equity benchmark, with both trade directions profitable and an 82.1% hit rate.

Component analysis attributes the performance to two elements derived from the fitted process — correct stop semantics and seasonal adjustment of the signal — while four further controls are shown to be neutral or harmful. A

stationary-bootstrap interval for the out-of-sample Sharpe, $[-0.160, 1.560]$, does not exclude zero: the relative and structural findings are robust, while the absolute level of risk-adjusted return is not resolvable on 625 out-of-sample days. Both statements form part of the result.

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A Reproducibility

A.1 Computational environment

Results were produced with Python 3.13 using `numpy`, `pandas`, `scipy`, `statsmodels` and `matplotlib`. The backtester is implemented directly rather than through a third-party framework, so that every accounting step — mark-to-market, cost application and position transition — is explicit and auditable.

A.2 Parameters

The search grid is $w \in \{20, 30, 40, 50, 60\}$ and $\theta_{\text{en}} \in \{1.5, 2.0, 2.5\}$, giving fifteen combinations. Fixed parameters are $\theta_{\text{ex}} = 0.5$, $\theta_{\text{st}} = 4.0$, time step 56 days, $\kappa = 0.01$, contract size 1,000 barrels, initial capital \$1,000,000, and risk-free rate 5% per annum. The harmonic order is $K = 1$ with a minimum of 504 observations before the first seasonal fit, refitted annually.

A.3 Determinism

All stochastic components are seeded: the bootstrap uses seed 7 and the scatter subsample in Figure 2(b) uses seed 0. Walk-forward folds are determined by index position and contain no randomness, so the entire pipeline is reproducible from the repository.